

Associate Architect with full-stack engineering experience specializing in **distributed real-time systems**, memory-optimized UI pipelines, and scalable cloud-native architecture. Proven track record of designing and shipping enterprise platforms orchestrating **8,000+ concurrent devices** across multi-tenant, multi-region deployments with **zero-downtime failovers**. Prolific builder and shipper — published apps, **agentic AI developer tools** with MCP integrations, and open-source projects across **Rust, Go, TypeScript, and Kotlin**. Known for **rapid end-to-end execution** from system design through production delivery at high velocity, while **mentoring engineers** and aligning cross-functional stakeholders.

SKILLS

< > LANGUAGES

C#
JavaScript / TypeScript
Go
Rust
Python
Java
C / C++
PHP
Dart

FRONTEND

React / Redux
Angular
HTML5 Canvas
Electron

DATABASES

MySQL
MongoDB
Redis
CouchDB / PouchDB
Neo4j GraphDB

CLOUD & INFRA

Azure
AWS
GCP
Nginx

MOBILE

Flutter
Kotlin / Android
React Native
Ionic

REALTIME & AI

WebSockets / SignalR
WebRTC
MQTT / IoT
Machine Learning
NLTK

... MORE

Node.js / Express
Agentic AI / MCP
Docker
CI / CD
Microservices
GraphQL

BLOG

medium.com/@dpkay

EDUCATION

B.E. in Computer Science & Engineering
VTU, Bengaluru — 2017

KEY PROJECTS

Video Surveillance System

2020 – Present

APRA LABS · ASSOCIATE ARCHITECT / LEAD

Architected an end-to-end cloud platform for orchestrating stable, **low-latency live streams**, configuration changes, and event captures for thousands of **geo-distributed cameras** across multiple tenants and regions. Engineered a scalable middleware layer with **localized brokers and edge protocols** for camera interfacing. Leveraged a **distributed service bus** for reliable cross-region message routing and implemented **real-time signaling and streaming state management** across cloud boundaries for multi-region tenant isolation and seamless failovers. Led and mentored a cross-functional team of 4 — developers, QA, and support — aligning stakeholders across engineering, operations, and client teams through development and multi-region production rollouts.

Impact: Achieved **zero-downtime regional failovers** and sub-second stream switching, scaling to **8,000+ concurrent cameras** while onboarding new enterprise tenants without additional infrastructure investment

Tech stack: C# (.NET), Azure Service Bus, SignalR, Azure Functions, Redis, WebRTC, MQTT, ONVIF, Milestone VMS, React, Redux, NSIS

Realtime Multi-purpose Dashboard

2023 – Present

APRA LABS · ASSOCIATE ARCHITECT / LEAD

As lead engineer, mentored a junior developer while designing and building a modular, content-heavy dashboard system **from scratch** — rendering hundreds of **live-updating widgets** and **millions of concurrent data points** on a complex, cross-building timeline without blocking the UI main thread. Engineered a **custom HTML5 Canvas rendering pipeline** bypassing traditional DOM limitations. Offloaded heavy calculations, stream parsing, and data manipulation to background **Web Workers**. Applied **progressive data fetching**, localized slice queries, and **lazy evaluation** of off-screen states to minimize memory overhead.

Impact: Sustained a **locked 60 FPS** across millions of live data points with **zero browser memory leaks** — powering 24/7 facility operations across multi-building security and access-control environments

Tech stack: C#, SignalR, SQL, Stored Procedures, React, Redux, HTML5 Canvas, Web Workers, TypeScript

E-Commerce Middleware & POS Platform

2017 – 2019

ALIENHU · SOFTWARE ENGINEER

Designed and developed a full-stack **B2B point-of-sale application** with a team, and **single-handedly built** the end-to-end **B2C e-commerce mobile app** and the shared sync middleware layer managing inventory updates, **real-time order notifications**, pricing changes, and **digital payment state flows** across **fragmented networks** where physical POS terminals frequently drop connectivity. Built a robust **transaction-handling and queuing mechanism** for reliable data syncing with safe **offline state management** to eliminate **race conditions** during POS resynchronization with the cloud backend.

Impact: Eliminated **double-selling and inventory lock conflicts** — enabling uninterrupted retail operations with **zero data loss** during offline-to-cloud resynchronization across fragmented POS networks

Features: Inventory management, GST management, **Tally integration**, **KOT and Bill customization**, **Pixel perfect printing**, Cart management, Online-pickup orders, **UPI payment**, OTP messaging

Tech stack: Node.js-Express, CouchDB-PouchDB, Angular, Electron, React Native, Ionic, WebSocket, NodeMailer

OPEN SOURCE & SHIPPED PRODUCTS

Terminal Studio

GPU-accelerated terminal multiplexer with built-in file browser, git diff viewer, and workspace persistence.

Rust, wgpu, egui

AfterCall Shield

Auto-dismisses post-call overlay pop-ups on Android. Published on Google Play Store.

Kotlin, Android

splitgame

Split-terminal game panel with Claude AI opponent via MCP integration. Published on npm.

TypeScript, Node.js

Also: gitreg (Rust) · claude-ticker (TypeScript) · tttag (Go) · TogglDev (Kotlin) · Workspace ID Pro (VS Code)

EXPERIENCE

Apra Labs

Associate Architect
2019 – Present

AlienHu

Software Engineer
2017 – 2019